

INTERNSHIP DATA MANAGEMENT SYSTEM

A PROJECT REPORT

for

Major Project-I (CA301P)

Session (2025-2026)

Submitted by

Mansi Garg

(202410116100116)

Manish Kumar Singh

(202410116100114)

**Submitted in partial fulfilment of the
Requirements for the Degree of**

MASTER OF COMPUTER APPLICATION

Under the Supervision of

Dr. Ankit Verma

Associate Professor



Submitted to

DEPARTMENT OF COMPUTER APPLICATIONS

KIET Group of Institutions, Ghaziabad

Uttar Pradesh-201206

(DECEMBER 2025)

CERTIFICATE

Certified that **Mansi Garg (202410116100116)**, **Manish Kumar Singh (202410116100114)** has/ have carried out the project work having “**Internship Data Management System**” (**Major-Project -I, CA301P**) for **Master of Computer Application** from KIET Group of Institutions (an Autonomous college) under Dr. A.P.J. Abdul Kalam Technical University (AKTU), Lucknow under my supervision. The project report embodies original work, and studies are carried out by the student himself/herself and the contents of the project report do not form the basis for the award of any other degree to the candidate or to anybody else from this or any other University/Institution.

Date:

MANSI GARG

(202410116100116)

MANISH KUMAR SINGH

(202410116100114)

This is to certify that the above statement made by the candidate is correct to the best of my knowledge.

Dr. Ankit Verma
Associate Professor
Department of Computer Applications
KIET Group of Institutions, Ghaziabad

Dr. Sachin Malhotra
Dean
Department of Computer Applications
KIET Group of Institutions, Ghaziabad

Internship Data Management System

Mansi Garg
Manish Kumar Singh

ABSTRACT

The Internship Data Management System is designed to streamline the entire process of managing internship-related activities within an academic institution. Internships play a crucial role in enhancing practical knowledge and industry exposure for students; however, the traditional manual approach to tracking student applications, company details, internship progress, and evaluation often leads to inefficiencies and data inconsistencies. This project provides a centralized and digital solution that simplifies the management of internship data for administrators, students, and companies.

The system enables administrators to maintain student records, manage company profiles, monitor internship opportunities, and track overall internship status. Students can upload resumes, view available internships, submit applications, and upload weekly or monthly progress reports. On the other hand, companies can post internship openings, review student applications, select candidates, and provide evaluations. The platform ensures transparent communication, timely updates, and easy accessibility of documents such as offer letters, reports, and certificates.

By implementing features like secure login, automated notifications, document management, and real-time status tracking, the system reduces manual effort and enhances accuracy. The digital workflow improves coordination among stakeholders, increases transparency, and ensures smooth execution of the internship process. Overall, the Internship Data Management System contributes to better organization, efficient monitoring, and improved decision-making, making it a valuable tool for educational institutions.

.

ACKNOWLEDGEMENT

Success in life is never attained single-handedly. I am deeply grateful to my project supervisor, **Dr. Ankit Verma**, for his invaluable guidance, unwavering support, and encouragement throughout my project work. His enlightening ideas, constructive comments, and thoughtful suggestions have greatly contributed to the completion of this project.

I would also like to extend my heartfelt thanks to **Dr. Sachin Malhotra**, Professor and Dean, Department of Computer Applications, for his insightful feedback and administrative support on various occasions, which proved to be immensely helpful during critical stages of the project.

I am fortunate to have many understanding friends who have supported me in numerous ways during challenging moments. Their assistance and companionship have been a constant source of motivation.

Finally, my sincere gratitude goes to my family members and all those who have directly or indirectly provided me with moral support, encouragement, and assistance. Their unwavering belief in me and their continuous efforts to keep my life filled with happiness and joy made the completion of this project possible.

Mansi Garg
Manish Kumar Singh

TABLE OF CONTENTS

S.NO.	TITLE	PAGE NO.
	Certificate	
	Abstract	
	Acknowledgement	
	Table of Contents	
	List of figures	
1.	CHAPTER 1: Introduction	
1.1	General	
1.2	Overview of the Internship Management System	
1.3	Objectives of the system	
1.3.1	Admin Benefits	
1.3.2	Student Benefits	
1.4	Problem Statement	
1.5	Target Audience	
1.6	Project Significance	
1.7	Limitations of the System	
2.	CHAPTER 2: FEASIBILITY STUDY	
2.1	Technical Feasibility	
2.2	Economic Feasibility	
2.3	Market Research	
2.4	Exisiting Solutions	
2.5	Gap Analysis	

3.	CHAPTER 3: PROJECT OBJECTIVE	
4.	CHAPTER 4: HARDWARE AND SOFTWARE REQUIREMENTS OF THE SYSTEM	
4.1	Hardware Requirements	
4.2	Software Requirements	
5.	CHAPTER 5: PROJECT FLOW	
5.1	Development Methodology	
5.2	Data Flow Diagram	
5.3	ER Diagram	
6.	CHAPTER 6: PROJECT OUTCOME	
6.1	System Features	
6.2	User Interface	
7.	REFERENCES	

LIST OF FIGURES

Fig. No.	Figure Name	Page No.
5.2.1	Data Flow Diagram – Level 0	
5.2.2	Data Flow Diagram – Level 1	
5.3.1	Student's View	
5.3.2	Teacher's/Admin View	
5.3.3	Functional Relationship	
6.2.1	Home Page	
6.2.2	Login Page	
6.2.3	Upload Section	
6.2.4	Review Section	

CHAPTER 1

INTRODUCTION

1.1 GENERAL

Internships bridge the gap between academic learning and real-world experience, helping students gain practical skills and industry exposure. Managing these internship activities manually often becomes difficult, especially when the number of students and companies increases. Paper-based records, scattered emails, and unorganized data lead to confusion and inefficiency.

The Internship Data Management System provides a centralized digital platform to simplify and organize all internship-related tasks. It allows students to view opportunities, apply, upload reports, and track their progress. Companies can post openings, review applications, and share feedback, while administrators can monitor the entire process and maintain accurate records.

Overall, the system improves communication, reduces manual work, ensures transparency, and makes internship management more efficient for all stakeholders.

1.2 OVERVIEW OF THE INTERNSHIP MANAGEMENT SYSTEM

The Internship Data Management System is designed to streamline the end-to-end process of managing internships in educational institutions. It provides a structured and efficient way to handle student applications, company details, internship postings, progress reports, and final evaluations. Instead of relying on manual records or scattered data, the system offers a centralized platform where all information is stored securely and can be accessed anytime. The platform includes separate modules for students, companies, and administrators, ensuring smooth interaction and clear roles. Students can apply for internships and upload necessary documents, companies can review applications and offer opportunities, and administrators can oversee the complete workflow and generate reports. With features like login authentication, automated notifications, and digital record-keeping, the system reduces paperwork, avoids duplication of data, and improves overall coordination.

In essence, the Internship Data Management System enhances efficiency, transparency, and convenience by simplifying the entire internship management process for all stakeholders involved.

1.3 OBJECTIVES OF THE SYSTEM

1.3.1 Admin Benefits

- To efficiently maintain and update student records in a centralized system.
- To manage company details and internship openings in an organized manner.
- To monitor student applications, selections, and internship progress.
- To reduce paperwork and simplify administrative tasks.
- To ensure accuracy, consistency, and easy retrieval of internship-related data.
- To generate reports and analytics for better decision-making.
- To provide smooth and clear communication between students and companies.
- To track the overall internship status for all students.

1.3.2 Student Benefits

- To easily access and explore available internship opportunities.
- To apply for internships through a user-friendly online platform.
- To upload required documents such as resumes, reports, and certificates.
- To track application status and internship progress in real-time.
- To receive timely notifications and updates from admin and companies.
- To keep all internship documents secure and organized in one place.
- To save time and reduce confusion by using a single integrated system.
- To improve communication with companies and internship coordinators.

1.4 PROBLEM STATEMENT

Managing internship activities manually becomes challenging as the number of students and companies increases. Traditional methods such as paper-based records, spreadsheets, and email communication often result in data duplication, miscommunication, delays, and lack of transparency. Tracking student applications, monitoring progress, collecting reports, and maintaining company details require extensive effort and can lead to errors or incomplete information. Additionally, students face difficulty accessing internship opportunities and staying updated about their application status, while administrators struggle to coordinate with multiple stakeholders simultaneously.

To address these issues, there is a need for a centralized, digital system that can store and manage all internship-related data efficiently. The system should streamline communication, automate updates, ensure data accuracy, and provide clear visibility of internship status for students, companies, and administrators. Such a solution would significantly reduce manual workload, improve coordination, and create a more organized and transparent internship management process.

1.5 TARGET AUDIENCE

The Internship Data Management System is designed to serve multiple stakeholders involved in the internship process. Its primary target audience includes:

1. Students

- Students who are actively looking for internships.
- Those who want to track their applications, upload reports, and stay updated on internship status.
- Students who need a centralized platform to manage all documents and communication.

2. Administrators / Internship Coordinators

- Faculty members or staff responsible for managing internship activities.
- Coordinators who handle student data, company information, and internship assignments.
- Admins who need to generate reports, monitor progress, and ensure smooth coordination.

3. Companies / Employers

- Organizations offering internship opportunities for students.
- Recruiters who want an easy way to post openings, review applications, and provide feedback.
- Companies seeking a structured platform to interact with students and administrators

1.6 PROJECT SIGNIFICANCE

The Internship Data Management System holds significant value for educational institutions, students, and companies involved in the internship process. By replacing manual and scattered methods with a centralized digital platform, the system enhances efficiency, organization, and transparency. It ensures that all internship-related activities—from application to completion—are managed smoothly and accurately.

For administrators, the system provides a reliable way to maintain student records, monitor ongoing internships, generate reports, and track progress without the need for excessive

paperwork. For students, it simplifies access to opportunities, allows easier submission of documents, and keeps them updated through clear status indicators. Companies also benefit from streamlined communication and structured application handling, improving the overall recruitment experience.

The system contributes to improved coordination among stakeholders, reduces the chances of data loss, and ensures timely decision-making. Additionally, by integrating data visualization and real-time tracking, it helps users understand trends and overall internship performance more effectively. Overall, the project significantly improves the quality of internship management, making the entire process more efficient, transparent, and user-friendly.

1.7 LIMITATIONS OF THE SYSTEM

Although the Internship Data Management System improves efficiency and organization, it still has certain limitations:

- 1. Internet Dependency**

The system requires a stable internet connection for accessing data, submitting applications, and uploading documents. Without internet, users cannot use most features.

- 2. Initial Setup Time**

Adding student records, company details, and past internship data into the system requires a significant amount of manual effort during the initial phase.

- 3. Technical Knowledge Requirement**

Users need basic computer and online navigation skills. New users may take time to adapt to the interface and features.

- 4. Limited Offline Functionality**

The system does not support offline operations, which can cause delays in uploading or retrieving information.

- 5. Security and Privacy Risks**

Even with protection measures, web-based systems may face potential cyber threats, data breaches, or unauthorized access if proper security is not maintained.

- 6. Maintenance Effort**

Regular updates, bug fixing, and server monitoring are required to ensure smooth functionality.

- 7. File Upload Constraints**

There may be limitations on file size, format, or storage capacity, which can sometimes restrict document uploads.

CHAPTER 2

FEASIBILITY STUDY

2.1 TECHNICAL FEASIBILITY

The Internship Data Management System is technically feasible as it can be developed using commonly available and reliable technologies. The system uses standard web development tools like HTML, CSS, JavaScript, and a suitable backend platform such as Node.js, PHP, or Python. Databases like MySQL or MongoDB provide secure and scalable storage for internship records.

The required hardware and software are easily accessible and do not demand high-end resources. The system can run on regular computers and can also be hosted on cloud servers for better performance. Since the chosen technologies are widely supported and well-documented, development, maintenance, and future updates are manageable.

Thus, the technical requirements for implementing the system are practical, affordable, and easy to fulfill.

2.2 ECONOMIC FEASIBILITY

The Internship Data Management System is economically feasible because it requires minimal investment for development and maintenance. The system can be built using open-source technologies, which reduces software licensing costs. Basic hardware such as standard computers and internet connectivity is sufficient, avoiding the need for expensive equipment.

Once implemented, the system helps reduce paper usage, manual workload, and administrative expenses. It also decreases the time spent on data management and communication, leading to improved productivity and cost efficiency. Long-term savings are achieved through reduced errors, faster processing, and easier data handling.

2.3 Market Research

Market research for the Internship Data Management System indicates a growing demand for organized and digital solutions in academic institutions. With the increasing number of students seeking internships and companies offering diverse opportunities, institutions require efficient systems to handle internship coordination. Many colleges and universities still rely on manual processes, which creates a gap that this system can effectively fill.

The shift toward digital transformation in education further supports the need for such platforms. Existing solutions in the market either focus on job portals or basic internship listings, but very few provide an integrated system that manages applications, communication, progress tracking, and document management in one place. This highlights an opportunity for a comprehensive internship management platform.

2.4 Existing Solutions

Several internship management platforms already exist, such as GoSprout, GradLeaders IMS, ImBlaze, and other educational management tools that help track student applications, maintain records, and connect students with companies. However, most of these solutions are either costly, complex, or focused on specific features rather than providing an integrated and customizable platform for educational institutions. Many institutions still rely on manual processes because existing systems may not fully suit their local needs, workflows, or budgets. Therefore, despite available tools, there remains a need for a simple, affordable, and user-friendly Internship Data Management System that can efficiently manage records, communication, and progress tracking in one centralized platform.

2.5 Gap Analysis

Although several internship management tools exist, most do not fully meet the specific needs of educational institutions. Many platforms are expensive, difficult to customize, or limited in features such as real-time tracking, document management, and communication between students, companies, and administrators. Institutions often struggle with mismatched workflows, lack of transparency, and dependency on manual data entry due to these shortcomings. The gap lies in the absence of a simple, affordable, and user-friendly system designed specifically for academic internship coordination. The proposed Internship Data Management System aims to fill this gap by offering centralized data handling, easy communication, streamlined application tracking, and improved accessibility for all stakeholders.

CHAPTER 3

PROJECT OBJECTIVE

The main objective of the **Internship Data Management System** is to develop a centralized, efficient, and user-friendly platform for managing all internship-related information. The system aims to streamline the entire process—from student registration and company details to application tracking, approval workflows, and performance evaluation—while ensuring accuracy, transparency, and easy accessibility. **Implementing Secure Login/Signup:** The system will provide a secure and seamless login/signup process using **Django's user authentication** system. Members will be able to create personalized accounts, log in securely, and reset their passwords when needed. The security features will include encryption, password hashing, and session management to protect user data.

- i. To centralize all internship-related data in a single digital platform.
- ii. To reduce manual paperwork and minimize errors.
- iii. To automate student registration and application tracking.
- iv. To maintain updated company profiles and internship opportunities.
- v. To enable smooth communication between students, coordinators, and companies.
- vi. To improve approval workflows (faculty/mentor verification).
- vii. To provide performance evaluation and feedback submission features.
- viii. To ensure secure storage and quick retrieval of records.
- ix. To generate reports and summaries for administrative decisions.
- x. To enhance transparency, efficiency, and overall data management.

CHAPTER 4

HARDWARE AND SOFTWARE REQUIREMENTS

4.1 Hardware Requirements

- A standard computer/laptop with minimum **2.0 GHz processor**
- **4 GB RAM** (minimum), **8 GB recommended**
- **500 GB storage** or more
- Stable internet connection
- Server machine (if deployed on local server)

4.2 Software Requirements

i. Frontend Technologies

- React.js
 - HTML5
 - CSS3 / Tailwind CSS
 - JavaScript (ES6+)
 - Axios / Fetch API (for API communication)
 - React Router (for navigation)
 - Redux / Context API (optional for state management).
-

ii. Backend Technologies

- Node.js
 - Express.js (for server-side routing)
 - REST API architecture
 - JWT / bcrypt (for authentication & security)
 - Nodemon (for development)
-

iii. **Database**

- MongoDB (preferred for MERN stack)
- Mongoose ORM (schema management)

CHAPTER 5

PROJECT FLOW

5.1 Development Methodology

The development of the Internship Data Management System follows the Agile Software Development Methodology. Agile allows the system to be developed in small, manageable increments. Each module is built, tested, and improved based on continuous feedback. This approach helps in maintaining flexibility, improves collaboration, and ensures better quality output.

Phases of Development

- 1. Requirement Analysis**

Requirements are gathered from users such as students, administrators, and coordinators. Functional and non-functional requirements are documented.

- 2. Planning**

The project is divided into smaller tasks or sprints. Each sprint focuses on specific features like registration, company management, or application tracking.

- 3. System Design**

System architecture, database schema, and user interface layout are designed. Flow diagrams and models are prepared for clarity.

- 4. Development**

Frontend components are developed using React. Backend APIs and logic are implemented using Node.js and Express. Database integration is done using MongoDB.

- 5. Testing**

Functionality testing, unit testing, and integration testing are performed. Bugs and errors are identified and resolved.

- 6. Deployment**

The system is deployed on a server or cloud platform for real-time use.

- 7. Maintenance**

Regular updates are made as per user feedback. Issues are fixed, and performance improvements are implemented.

5.2 Data flow Diagram

LEVEL 0:

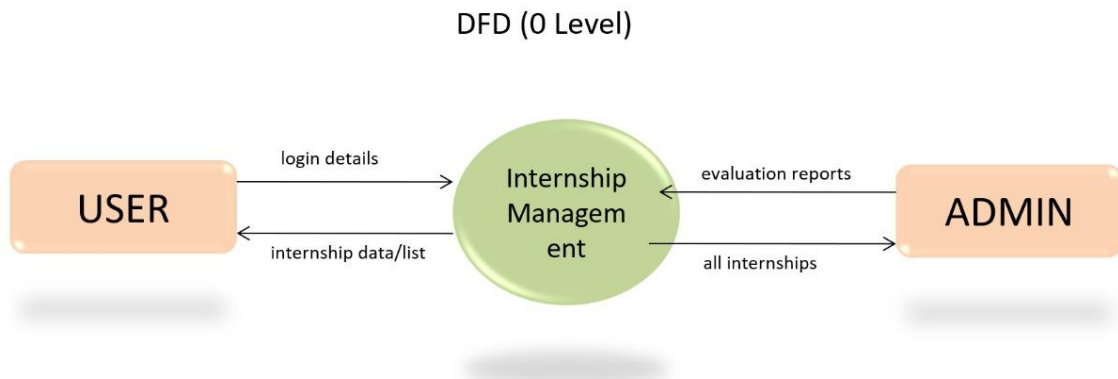


Figure-5.2.1

LEVEL 1:

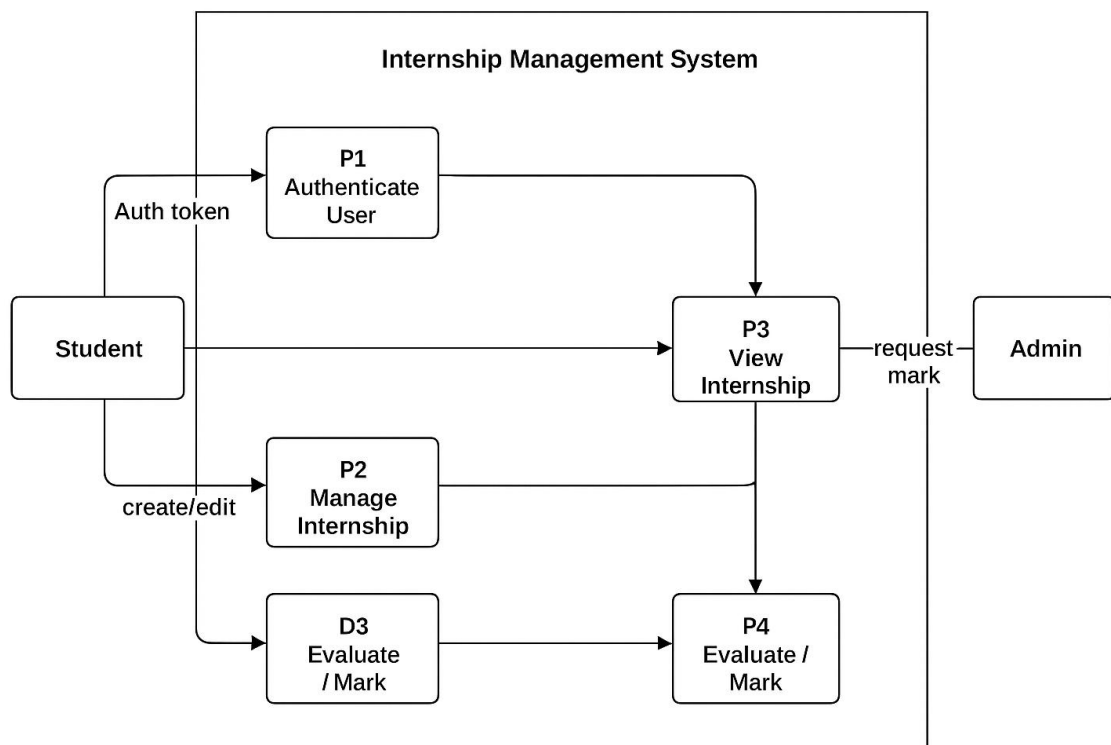


Figure-5.2.2

5.3 ER Diagram

Student's view:

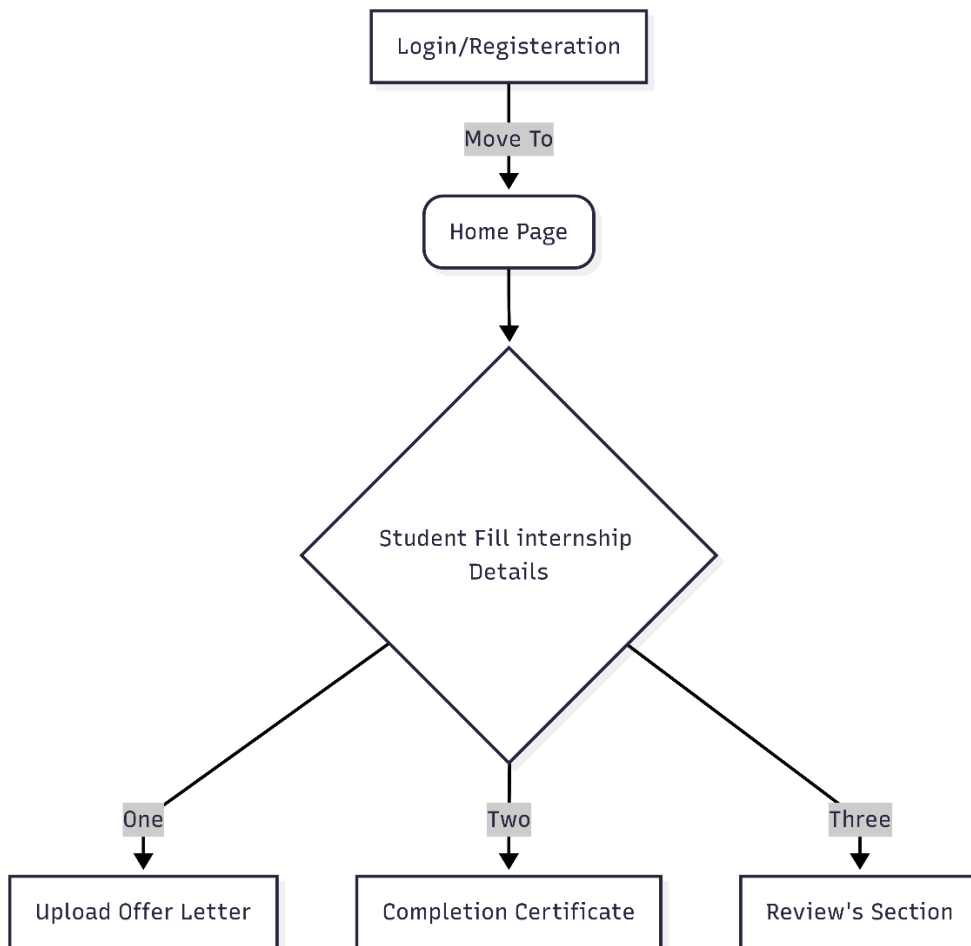


Fig 5.3.1

Teacher/ Admin View:

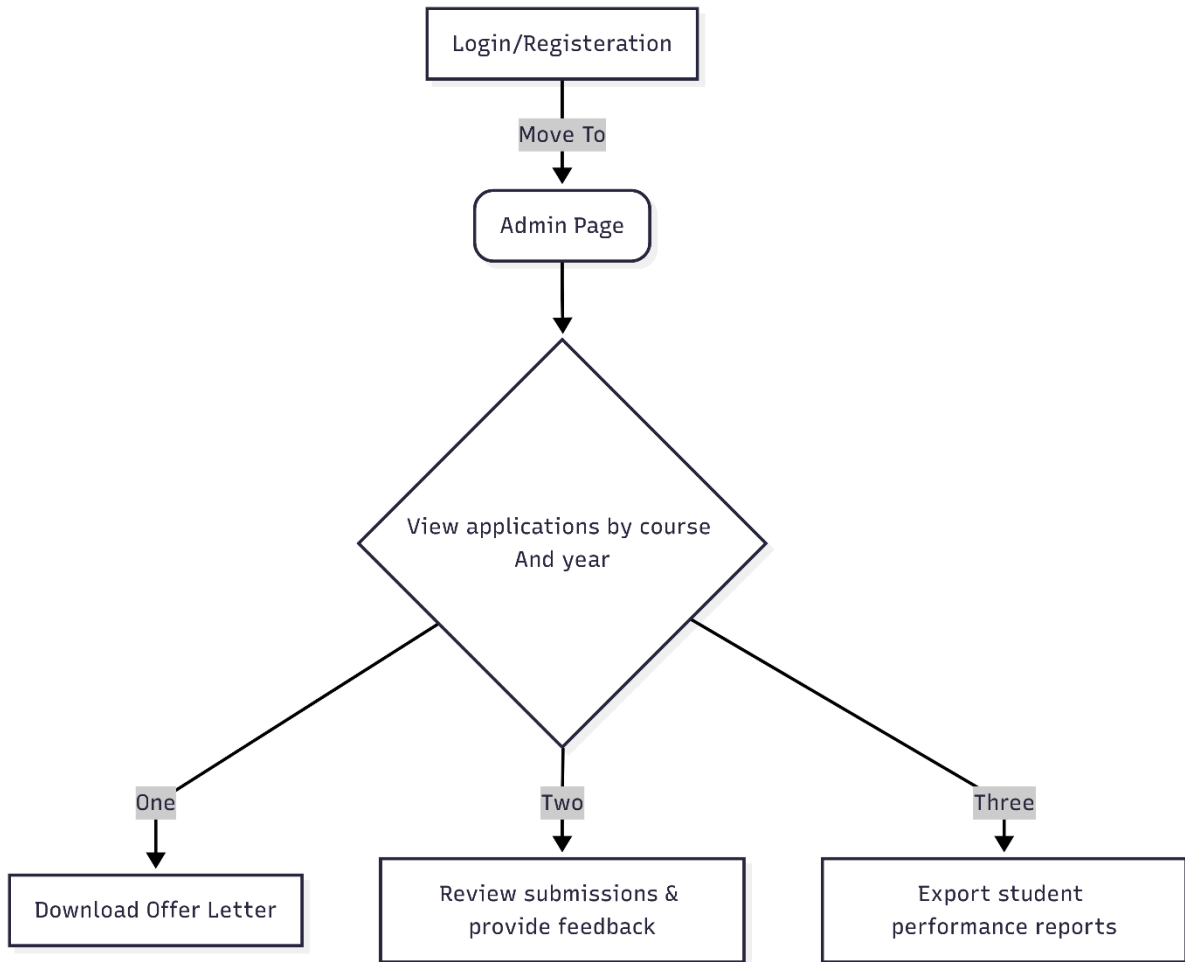


Fig. 5.3.2

FUNCTIONAL DEPENDENCY:

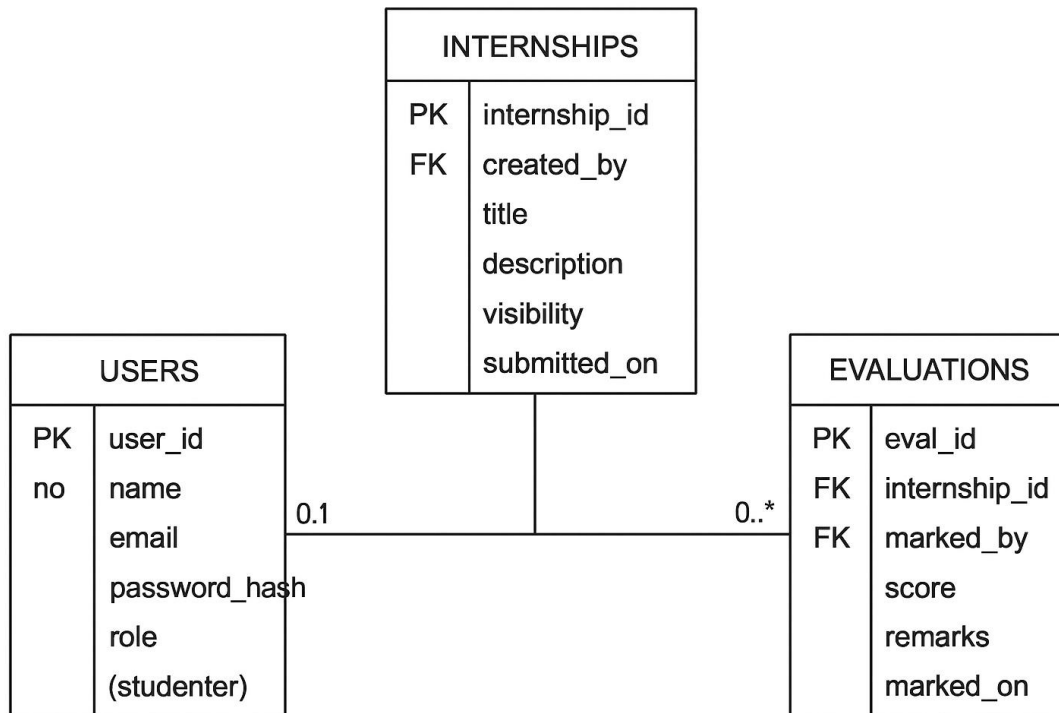


Figure – 5.3.4

CHAPTER 6

PROJECT OUTCOME

6.1 System Features

- **Secure Login/Signup:**
Ensures only authorized users can access the system, keeping intern and company data safe.
- **Intern Profiles and Records:**
Stores details of interns, including personal info, assigned projects, attendance, and performance.
- **Project Tracking and Reports:**
Allows tracking of intern tasks, deadlines, and progress, and generates reports for supervisors.
- **Notifications and Updates:**
Sends alerts for deadlines, approvals, and important announcements to keep interns and managers informed.

6.2 User Interface

- **Overview:**
Intuitive, easy-to-use interface with clear navigation and minimal learning curve.
- **Navigation and Layout:**
Dashboards, menus, and buttons are organized for quick access to intern data, projects, and reports.
- **Responsive Design:**
Accessible on desktop and mobile devices for supervisors and interns.
- **Interactive Features:**
Includes search, filters, and notifications for easy data management and timely updates.

HOME PAGE:

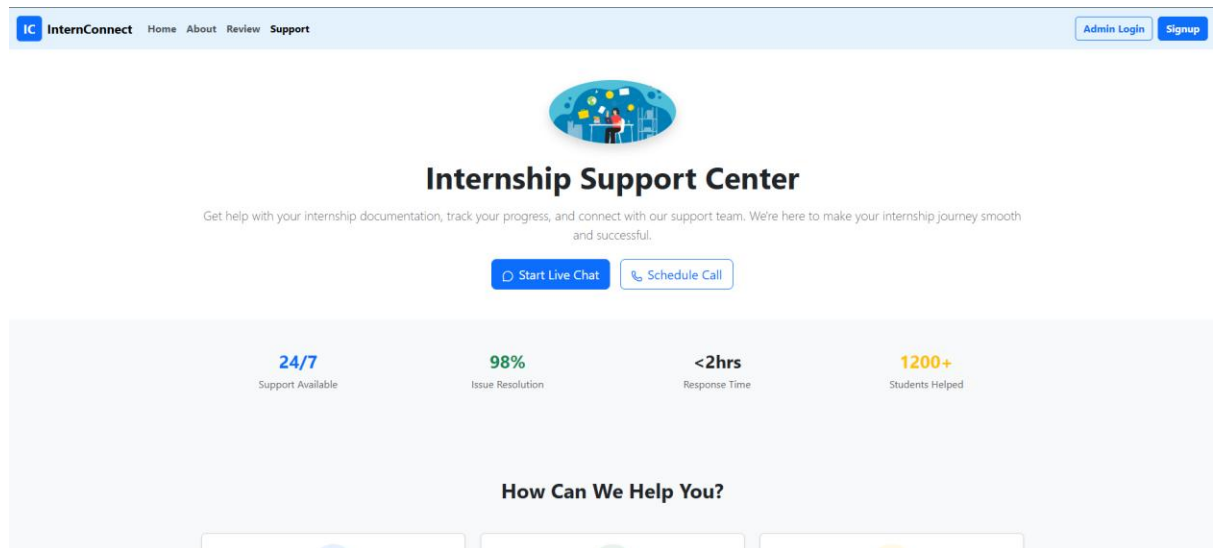


Fig 6.2.1

LOGIN:

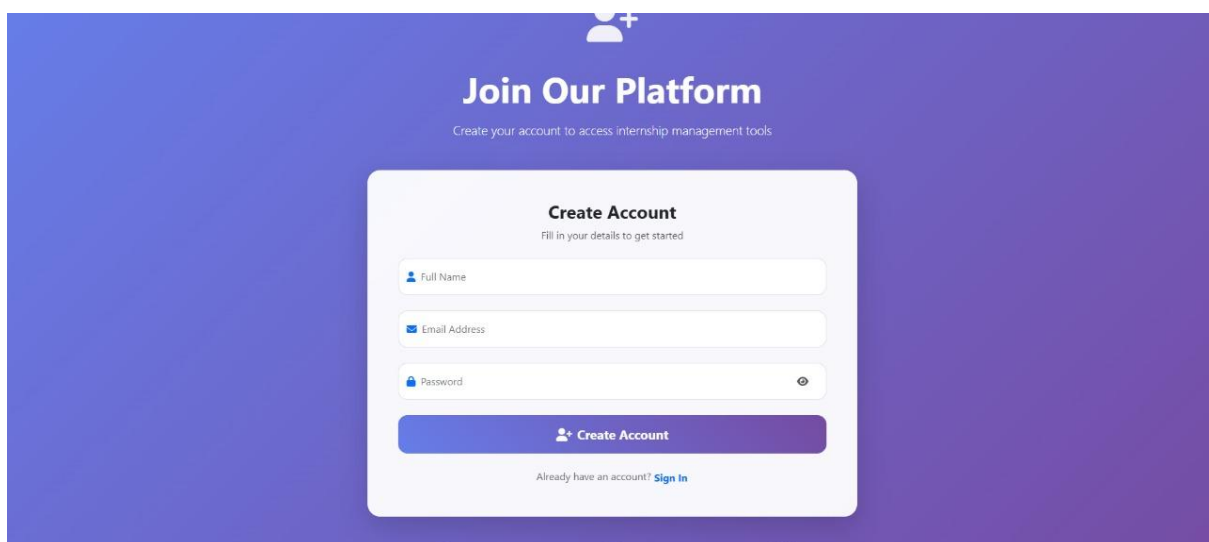


Fig 6.2.2

UPLOAD SECTION:

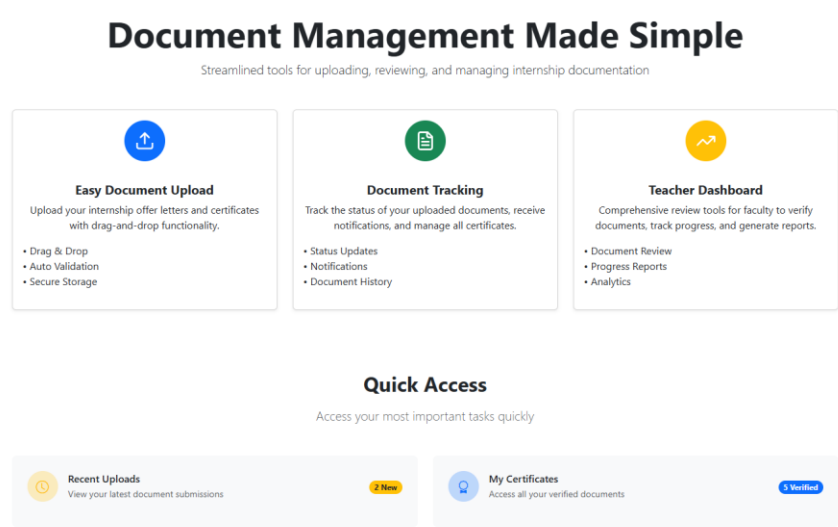


Fig 6.2.3

REVIEW SECTION:

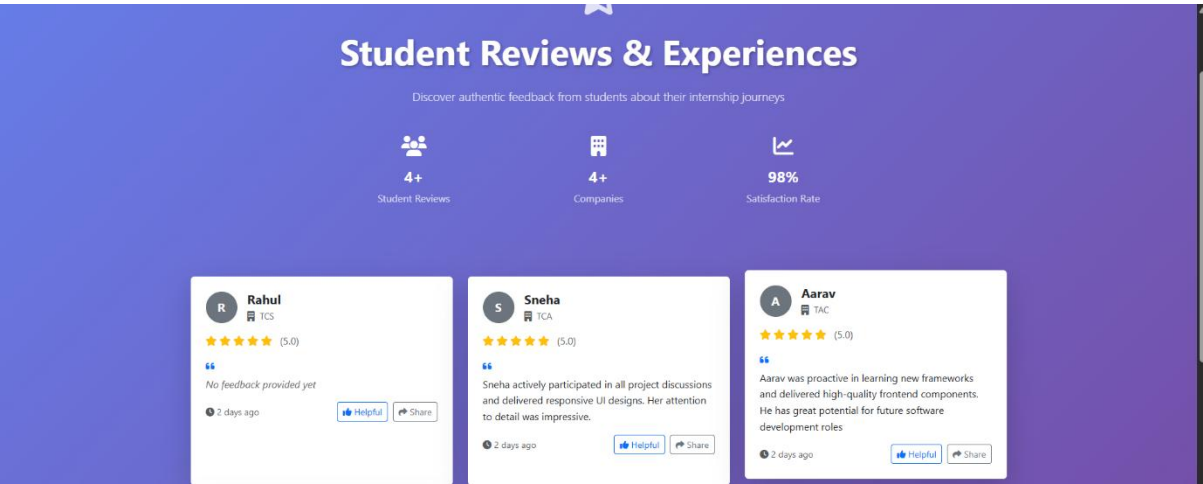


Fig 6.2.4

REFERENCES

1. M. G. Salazar and J. R. Lopez, "Design and development of a gym management system," Proceedings of the International Conference on Computing and Information Technology, vol. 4, no. 2, pp. 33-38, 2021.
2. K. J. M. Hartmann, "Web-based gym management system using Django," International Journal of Computer Applications, vol. 150, no. 1, pp. 67-72, 2016.
3. "HTML: Hypertext Markup Language," MDN Web Docs. [Online]. Available: <https://developer.mozilla.org/en-US/docs/Web/HTML>. [Accessed: Dec. 14, 2024].
4. "CSS: Cascading Style Sheets," MDN Web Docs. [Online]. Available: <https://developer.mozilla.org/en-US/docs/Web/CSS>. [Accessed: Dec. 14, 2024].
5. "JavaScript: Programming language for the web," MDN Web Docs. [Online]. Available: <https://developer.mozilla.org/en-US/docs/Web/JavaScript>. [Accessed: Dec. 14, 2024].
6. D. S. Lee, "Django web framework for building modern applications," Proceedings of the International Conference on Software Engineering and Development, vol. 2, pp. 150-155, 2020.
7. S. Kumar, A. Sharma, and R. Gupta, "Integration of payment gateway in gym management systems," Journal of Computer Science and Technology, vol. 7, no. 5, pp. 112-118, 2018.
8. "Stripe API documentation," Stripe. [Online]. Available: <https://stripe.com/docs/api>. [Accessed: Dec. 14, 2024].
9. "PayPal REST API," PayPal Developer. [Online]. Available: <https://developer.paypal.com/docs/api/overview/>. [Accessed: Dec. 14, 2024].
10. R. M. Patel, "Real-time data analytics for gym management," International Journal of Data Science and Analytics, vol. 5, no. 3, pp. 42-47, 2019.